

TECHNICAL REFERENCE

Shokunin 職人 AI Ecosystem

37 skills, ChromaDB memory v4.1, MCP servers, subagents, and Windows automation — an open source AI engineering ecosystem for OpenCode.

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github.com/EliasOulkadi/shokunin

| Contents

01	Introduction
02	Architecture Overview
03	The 37 Skills
04	Memory System v4.1
05	MCP Servers
06	Subagents
07	Scripts & Automation
08	Configuration
09	CI/CD & Testing
10	Quick Reference
11	Comparison & Roadmap

| Introduction

Shokunin (職人, Japanese for artisan) turns OpenCode into a production-grade AI engineering workstation with 37 specialized skills, persistent ChromaDB memory, and weekly automations — all open source and free.

Instead of starting every session with a blank-slate agent, Shokunin provides a complete engineering environment: domain expertise through auto-activating skills, persistent context across sessions through ChromaDB memory, direct system access through MCP servers, and automated maintenance through PowerShell scripts and Windows Task Scheduler.

Everything runs on your machine. No cloud dependencies, no telemetry, no subscriptions. The AI runs on NVIDIA's free API tier or Ollama locally.

One command install:

```
irm https://raw.githubusercontent.com/EliasOulkadi/shokunin/master/install.ps1 | iex
```

Core Principles

- Zero cost — NVIDIA free API, open source, no subscriptions
- Local-first — all data stays on your machine
- Auto-activating — skills match your task description, no manual selection
- Persistent memory — session context survives across restarts
- Idempotent installer — safe to re-run, creates backups before changes

Architecture Overview

Five independent layers that each serve a specific role. Every layer can be replaced or extended without affecting the others.

Layers

Layer	Components	Purpose
Interface	OpenCode, VS Code + Cline, WezTerm	Terminal and editor entry points
Agent	37 skills, superpowers plugin, 4 subagents	Domain expertise for any task
Infrastructure	3 MCP servers, ChromaDB	Filesystem, web fetch, persistent memory
AI Providers	NVIDIA Kimi K2.6, Ollama local, OpenAI-compatible	Flexible model backend
System	PowerShell profile, Task Scheduler, scripts	Automation and maintenance

OpenCode · VS Code + Cline · WezTerm
37 skills · superpowers plugin · 4 subagents
MCP: filesystem · fetch · memory (ChromaDB)
NVIDIA · OpenAI · Anthropic · Ollama (any model)
PowerShell profile · Task Scheduler · 8 scripts

Data Flow

When you describe a task, OpenCode reads the name and description from every skill. When a match is found, the full SKILL.md instructions load into context. The agent uses MCP servers for file access, web fetching, and memory. At session end, the wrapper captures the console buffer and saves it to both ChromaDB (semantic) and markdown (grep-able).

| The 37 Skills

Every skill is a SKILL.md file that teaches the agent a domain. They auto-activate based on what you ask. Each includes a procedural workflow, error handling table, production checklist, and anti-patterns.

Infrastructure (5)

Skill	Description	Scripts
docker	Multi-stage builds, BuildKit cache, compose watch, vulnerability scanning	optimize-dockerfile, scan-image
kubernetes	Gateway API, service mesh, eBPF observability, HPA, pod debugging	debug-pod, generate-manifest
terraform	Stacks, test framework, pre/postconditions, state surgery	validate-all, plan-env
ci-cd	GitHub Actions, GitLab CI, CircleCI, OIDC auth, canary deployments	generate-pipeline
db-admin	PostgreSQL backup with verification, replication, WAL archiving, PITR	backup-db, monitor-db

Backend (4)

Skill	Description
auth-architect	OAuth2 + PKCE, WebAuthn/Passkeys, JWT rotation, RBAC/ABAC, OWASP Top 10
api-forge	REST/GraphQL APIs, OpenAPI 3.1, webhooks with HMAC, idempotency keys
db-sculptor	Index strategy, EXPLAIN ANALYZE, expand/contract migrations, Prisma/Drizzle
error-handler	OpenTelemetry tracing, retry with jitter, circuit breaker, error budgets

Frontend (5)

Skill	Description
component-forge	React/Vue/Svelte components, TypeScript strict, WCAG 2.2, compound components
responsive-engine	Container Queries, clamp(), :has(), subgrid, dvh/svh units
motion-craft	Web Animations API, Scroll-Driven Animations, FLIP, spring physics
landing-craft	CRO frameworks, A/B testing, form optimization, Core Web Vitals
ui-ux-pro-max	DB of UI patterns, color palettes, font pairings across 13 stacks

Mobile (2)

Skill	Description
flutter	Riverpod, Impeller, Dart 3.7+, Pigeon, Clean Architecture
react-native	Expo Router, FlashList, Reanimated 4, New Architecture, Hermes

Quality (2)

Skill	Description
test-commander	Testing Trophy methodology, Vitest, MSW, Playwright E2E, visual regression
performance-profiler	Lighthouse, Core Web Vitals, bundle analysis, caching strategies

Content & Business (5)

Skill	Description
communication	Professional emails, SBI/BID feedback, difficult conversations, meeting notes
content-marketing	AIDA/PAS/BAB frameworks, headline formulas, Cialdini psychology
business-proposals	Cold email sequences, Good-Better-Best pricing, SOWs, pitch decks
seo-geo	SEO + GEO for 2026, llms.txt, schema.org, AI Overviews optimization
translate-craft	8 languages, cultural adaptation, ICU syntax, RTL layout

Productivity (9)

Skill	Description
git-workflow	Branch/commit/PR/rebase automation with 4 PowerShell scripts
windows-powershell	System info, disk cleanup, dev tools install, PS profile
strategy	Brainstorming, decision frameworks (RICE, weighted scoring)
design	Brand guidelines, design tokens, visual briefs
documentation	READMEs, API docs, changelogs, knowledge base articles
runbook-gen	Incident runbooks, post-mortems, war room setup
finance	5-pillar planning, tax optimization, FIRE, insurance
legal-counsel	GDPR, EU AI Act, CCPA, HIPAA, DMCA reference
whendone-plus	Desktop notifications when long commands finish

Documents (2)

Skill	Description
kami	Professional PDF generation with parchment design, 8 templates
portfolio-auto	Auto-sync GitHub repos to portfolio website

System (3)

Skill	Description
memory	ChromaDB persistent memory, search, save, session management
chromadb	Manage vector DB: view, search, delete, backup, reset
documentation	READMEs, API docs, changelogs, knowledge bases

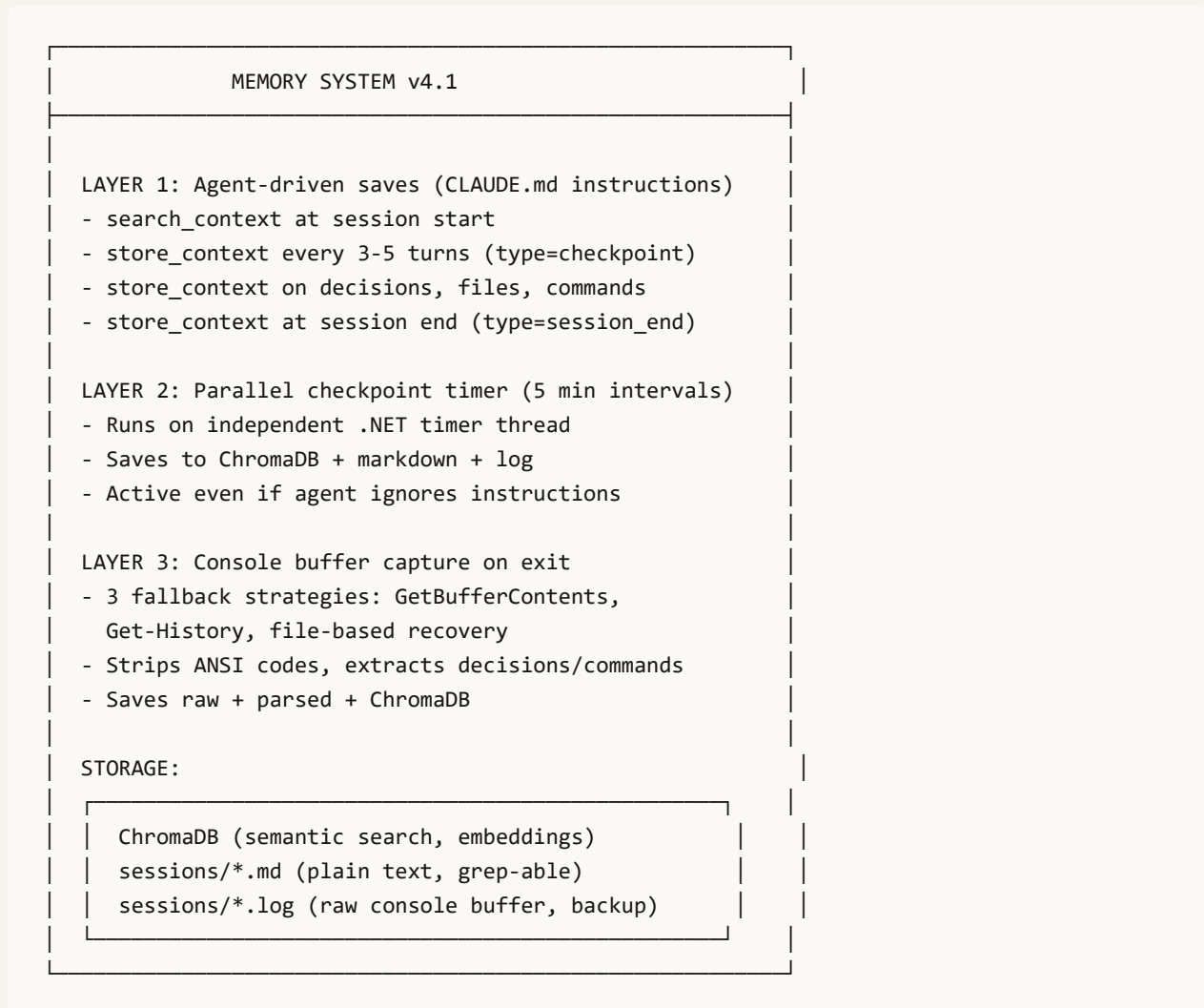
Skill Format

Every v3+ skill includes: YAML frontmatter with trigger-optimized description and negative triggers, numbered procedural workflow, error handling table (scenario → cause → fix), production checklist (verifiable items before marking done), anti-patterns table (what not to do with concrete fixes), and cited sources (OWASP, NIST, PostgreSQL docs, MDN).

Memory System v4.1

Three layers ensure no context is lost between sessions. ChromaDB provides semantic search; a parallel checkpoint timer saves every 5 minutes; the wrapper captures the console buffer on exit.

Architecture



Storage

What	Path	Notes
Vector database	~/.shokunin/memory/chroma_db/	400 KB base + ~1 KB per entry
Session logs	~/.shokunin/memory/sessions/	Markdown + JSON + raw logs
Active session	~/.shokunin/current-session.json	Readable by agent via filesystem MCP
Weekly backups	~/.shokunin/backups/	Zip archive, Sunday 21:00
MCP server log	~/.shokunin/memory/mcp-server.log	All operations logged

Agent Environment Variables

Written by `run-opencode.ps1` before launching OpenCode:

Variable	Value
SHOKUNIN_SESSION_ID	Current session ID (e.g. session-20260514-103000-A7F3)
SHOKUNIN_PROJECT	Current working directory at session start
SHOKUNIN_MCP_HEALTHY	"1" if MCP server responded, "0" otherwise
SHOKUNIN_LAST_SESSION	Set after session ends (for script use)

Data Format

Each memory entry stores structured metadata with a type field for precise filtering:

```
{
  "type": "decision | file | command | preference | checkpoint | session_end",
  "timestamp": "ISO-8601",
  "project": "project-name",
  "tags": ["tag1", "tag2"],
  "session_id": "session-YYYYMMDD-HHMMSS-NNNN",
  "text": "Context-rich description"
}
```

Fallback Chain

- 1. store_context via MCP server (ChromaDB) — primary
- 2. chroma-helper.py save (Python ChromaDB direct) — retry 2x
- 3. sessions/*.md — markdown file on disk
- 4. CLAUDE.md fallback: Python script via chroma-helper.py

| MCP Servers

MCP (Model Context Protocol) servers give OpenCode access to external tools. They start on demand via `opencode.json` and consume no resources when idle.

Server	Tools	Purpose
filesystem	read, write, edit, search, list	Safe file operations with path validation (root: C:\Users\swagger)
fetch	fetch, fetch_json	Download web pages and APIs in real time
memory	store_context, search_context, get_session_summary	ChromaDB vector memory with type filtering

Memory MCP Server

150 lines of Python, JSON-RPC 2.0 over `stdin/stdout`. Commands:

```
echo '{"jsonrpc":"2.0","id":1,"method":"tools/call",
      "params":{"name":"store_context",
                "arguments":{"text":"...", "session_id":"...",
                           "type":"decision", "tags":["auth"]}}}' | python mcp-server.py
```

| Subagents

Four subagents handle specialized tasks. Each runs with its own model and focus area, configured in `opencode.json`.

Subagent	Model	Fallback	Purpose
code-review	NVIDIA Kimi K2.6	Ollama Qwen3 14B	Analyze diffs for bugs, security, code smells
code-review-local	Ollama Qwen3 14B	—	Offline code review
debugger	NVIDIA Kimi K2.6	Ollama Qwen3 14B	Root cause analysis from stack traces
debugger-local	Ollama Qwen3 14B	—	Offline debugging

Scripts & Automation

Eight PowerShell and Python scripts handle session management, memory operations, health checks, and weekly maintenance.

Script Inventory

Script	Language	Purpose
run-opencode.ps1	PowerShell	Main wrapper: env vars, MCP check, checkpoint timer, buffer capture, ChromaDB save
chroma-helper.py	Python	Direct ChromaDB access: save, search, count (used by all PS scripts)
save-memory.ps1	PowerShell	Saves with retries (2x), falls back to markdown
search-memory.ps1	PowerShell	Unified search: ChromaDB + grep, dedup, relevance sort
read-transcript.ps1	PowerShell	Strips ANSI, extracts decisions and commands, generates markdown
memory-healthcheck.ps1	PowerShell	Validates all components: ChromaDB, MCP, scripts, config
test-memory.ps1	PowerShell	Comprehensive automated tests (12 checks)
weekly-healthcheck.ps1	PowerShell	Temp cleanup, DB backup, disk space check (Task Scheduler, Sunday 21:00)

Wrapper Session Flow

When the user types `opencode` , the PowerShell profile shadows it to run `run-opencode.ps1` :

1. Generate session ID and write to `current-session.json`
2. Set environment variables for the agent
3. Ping MCP server for health status
4. Increase console buffer to 9999 lines
5. Start checkpoint timer (fires every 5 minutes)
6. Launch real opencode binary
7. On exit: stop timer, read buffer, try 3 capture strategies
8. Save raw log + parsed markdown + ChromaDB
9. Restore original console buffer size

Configuration Files

opencode.json

MCP server config, provider setup (NVIDIA + Ollama), subagent definitions, superpowers plugin.
Located at `~/.config/opencode/opencode.json`.

CLAUDE.md

Global AI instructions with mandatory memory section. Located at `~/.claude/CLAUDE.md`. Contains identity, language, code rules, scope rules, and the full memory system protocol (search, save, checkpoint, session end, MCP fallback).

AGENTS.md

Agent instructions with memory protocol. Located at `~/.claude/AGENTS.md`.

PowerShell Profile

20+ aliases (git, npm, docker, utils), PSReadLine autocomplete, oh-my-posh prompt, Telegram bot auto-start, opencode shadow function. Located at `$PROFILE`.

Other Configs

File	Purpose
<code>.wezterm.lua</code>	GPU terminal with Catppuccin theme
<code>bookmarklet.html</code>	Send web pages to OpenCode
<code>dashboard.html</code>	Local ecosystem status viewer
<code>install.ps1</code>	517-line idempotent installer

| CI/CD & Testing

GitHub Actions validates all memory components on every push to master.

CI Pipeline

Three parallel jobs (total ~2 minutes):

Job	OS	Checks
test-python	Ubuntu	ChromaDB import, chroma-helper save/search, MCP tools/list, MCP store/search
test-scripts	Windows	PowerShell script syntax validation, chroma-helper Windows test
validate-config	Ubuntu	JSON validity, SKILL.md required fields, CLAUDE.md memory section

The pipeline triggers on pushes to master and pull requests that touch `.pack/memory/` , `.pack/scripts/` , `.pack/CLAUDE.md` , OR `install.ps1` .

Quick Reference

Key Locations

What	Path
Skills	~/.config/opencode/skills/
OpenCode config	~/.config/opencode/opencode.json
CLAUDE.md	~/.claude/CLAUDE.md
AGENTS.md	~/.claude/AGENTS.md
Memory data	~/.shokunin/memory/
Current session	~/.shokunin/current-session.json
Session logs	~/.shokunin/memory/sessions/
Backups	~/.shokunin/backups/
PowerShell profile	\$PROFILE
GitHub repo	github.com/EliasOulkadi/shokunin

Startup Commands

Action	Command
Start session (memory capture)	opencode (or .\run-opencode.ps1)
Test memory system	.\test-memory.ps1
Run healthcheck	.\memory-healthcheck.ps1
Git status	gst
Git add + commit	ga + gc "message"
npm install	ni
npm run dev	nrd

Memory Queries

Goal	Command
Search past sessions	<code>search-memory.ps1 -Query "auth flow" -Project "myapp"</code>
Save a note	<code>save-memory.ps1 -Text "..." -Tags decision,auth -Project "myapp"</code>

Comparison & Roadmap

How Shokunin compares to other AI skill repositories and what is planned for future releases.

Feature Comparison

Feature	Shokunin	superpowers	claude-skills
Skills	37 curated	14 methodology	254 scripts
Memory system	ChromaDB v4.1 ✓	X	X
MCP servers	3 included ✓	X	X
One-command installer	✓	Plugin-based	X
Subagents (4)	✓	✓	X
CI/CD pipeline	GitHub Actions ✓	✓	X
Auto-save wrapper	Console capture ✓	X	X
Checkpoint timer	5 min intervals ✓	X	X
Multi-model support	NVIDIA + Ollama + API ✓	Claude only	Claude only
Cost	Zero	Zero	Zero

Version History

Version	Key Changes	Date
v1.0	Initial 29 skills	2025.Q4
v2.0	Overhaul: all skills rewritten	2026.Q1
v3.0	12 skills upgraded, memory MCP, subagents, superpowers plugin	2026.04
v4.0	Memory rewrite: structured data, run-opencode.ps1, chroma-helper.py	2026.05.14
v4.1	Checkpoint timer, CI pipeline, profile shadow, 3 capture strategies	2026.05.14

Roadmap

- v5.0: Cross-platform support (bash scripts for macOS/Linux)

- v5.1: Visual memory dashboard with web UI
- v5.2: Memory export/import for sharing and team collaboration
- Future: OpenCode plugin for IDE integration, community skill marketplace

Links

Repo: github.com/EliasOulkadi/shokunin

Install: `irm https://raw.githubusercontent.com/EliasOulkadi/shokunin/master/install.ps1 | iex`

License: MIT

PDF Guides: github.com/EliasOulkadi/shokunin/blob/master/docs/